AI-Lingo

User Stories & Use Cases

Scenarios Illustrating How Users Interact with the System

Project: AI-powered English Language Learning Application

User Stories

US-01 – Create an Account

As a new user, I want to create an account so that I can save my learning data and track my progress.

US-02 – Log In

As a registered user, I want to log in securely so that I can access my personal learning information.

US-03 – Choose English Level

As a learner, I want to select my English level so that the application can adapt conversations to my ability.

US-04 – Assess English Level

As a learner, I want to complete an optional AI-based level assessment so that the application can estimate my English proficiency.

US-05 – Select a Scenario

As a learner, I want to choose a real-world scenario, such as a job interview, restaurant, airport, hotel, shopping, or business meeting, so that I can practice useful English.

US-06 – Practice with the AI Coach

As a learner, I want to have an interactive conversation with an AI coach so that I can improve my communication skills through realistic practice.

US-07 – Receive Conversation Feedback

As a learner, I want to receive a score, corrections, and recommendations after a conversation so that I can understand and improve my mistakes.

US-08 – Save New Vocabulary

As a learner, I want to review and save new words, meanings, and examples from previous conversations so that I can build my vocabulary.

US-09 – Track Learning Progress

As a learner, I want to view my previous conversations, scores, and vocabulary growth so that I can monitor my learning over time.

US-10 – Assess Learning Progress

As a learner, I want to take periodic English assessments and compare my results with previous assessments so that I can measure my improvement and identify areas that need more practice.

US-11 – Generate a Personalized Learning Plan

As a learner, I want the AI coach to create a personalized learning plan based on my English level, assessment results, strengths, and weaknesses so that I can improve through structured practice.

US-12 – Update the Learning Plan

As a learner, I want my learning plan to be updated based on my latest performance so that my learning activities match my current level and needs.

US-13 – Manage Profile and Settings

As a learner, I want to manage my profile and application settings so that I can keep my account information and preferences up to date.

Use Cases

Each use case describes the actor, goal, preconditions, main flow, alternative flow, and postconditions.

UC-01 – Register and Log In

Actor: New or registered user

Goal: Access a personal AI-Lingo account

Preconditions: The application is installed and an internet connection is available.

Main Flow:

1. The user opens AI-Lingo.
2. The user chooses Register or Log In.
3. For registration, the user enters the required details; for login, the user enters their credentials.
4. Firebase Authentication validates the request.
5. The application opens the user's home screen when authentication succeeds.

Alternative Flow:

If the details are invalid or authentication fails, the application displays an error and allows the user to try again.

Postconditions:

The user is authenticated and can access their own learning data.

UC-02 – Choose or Assess English Level

Actor: Learner

Goal: Set an appropriate English level

Preconditions: The user is logged in.

Main Flow:

6. The user opens the onboarding or level section.
7. The user selects a level or chooses the optional AI assessment.
8. If assessment is selected, the application presents English questions.
9. The user submits answers and Gemini estimates the user's level.
10. The application displays and saves the selected or estimated level.

Alternative Flow:

If the user does not complete the assessment, the user can choose a level manually or return later, depending on the implemented design.

Postconditions:

A level is available to guide the difficulty of future practice.

UC-03 – Practice an AI Conversation

Actor: Learner

Goal: Practice English in a realistic situation

Preconditions: The user is logged in and has a selected or estimated English level.

Main Flow:

11. The user selects a scenario, such as Job Interview or Restaurant.
12. The user starts a conversation.
13. The application sends the level and scenario instructions to Gemini through Firebase AI Logic.
14. The AI coach responds with a suitable opening message.
15. The user and AI exchange messages while the conversation context is maintained.
16. The user ends the session.

Alternative Flow:

If the AI request fails or connectivity is unavailable, the application displays an appropriate error and offers a retry when possible.

Postconditions:

The conversation is ready for evaluation and can be saved to the user's history.

UC-04 – Evaluate a Completed Conversation

Actor: Learner

Goal: Understand performance and correct mistakes

Preconditions: A conversation transcript is available.

Main Flow:

17. The user finishes a conversation.
18. The application submits the transcript to Gemini with evaluation instructions.
19. The AI evaluates grammar, vocabulary, fluency, and overall performance.
20. The application displays a score, mistakes, corrected sentences, new vocabulary, and recommendations.
21. The evaluation and related learning data are saved in Firestore.

Alternative Flow:

If evaluation fails, the application informs the user and allows the evaluation to be retried.

Postconditions:

The user can review feedback and use it in future practice.

UC-05 – Review and Save Vocabulary

Actor: Learner

Goal: Build a personal collection of useful words

Preconditions: Vocabulary from previous conversations or lessons is available.

Main Flow:

22. The user opens the Vocabulary section.
23. The application displays saved words, meanings, examples, and learning status.
24. The user reviews a word and its example.
25. The user saves or updates the word's learning status, where supported.

Alternative Flow:

If there are no saved words yet, the application displays an empty-state message and suggests practicing a conversation.

Postconditions: The user's vocabulary collection is updated and available for later review.

UC-06 – View Learning Progress

Actor: Learner

Goal: Review historical learning activity

Preconditions: The user is logged in and learning data is available.

Main Flow:

26. The user opens the Progress section or home dashboard.
27. The application retrieves available progress data.
28. The application displays previous conversations, performance scores, conversation counts, and vocabulary growth.
29. The user reviews the information to identify areas for improvement.

Alternative Flow:

If the user has no history, the application explains that progress will appear after the first learning activities.

Postconditions:

The user can monitor recorded learning activity over time.

UC-07 – Assess Learning Progress

Actor: Learner

Goal: Measure improvement between assessments

Preconditions: The user is logged in. A previous assessment is helpful for comparison but is not required for the first assessment.

Main Flow:

30. The user opens the Progress Assessment section.
31. The application provides an English assessment suitable for the user's level.
32. The user answers and submits the questions.
33. The AI evaluates the assessed language skills and returns structured results.
34. The application displays the current result.
35. If previous results exist, the application compares them with the current result and highlights changes and areas needing practice.
36. The application saves the assessment result for future comparison.

Alternative Flow:

If no previous assessment exists, the current result becomes the baseline. If an assessment cannot be completed, the application explains the issue and allows a retry.

Postconditions:

The user can review assessment history and see evidence of progress over time.

UC-08 – Generate a Personalized Learning Plan

Actor: Learner

Goal: Receive a structured plan tailored to learning needs

Preconditions: The user is logged in. A selected level, assessment results, or previous learning data should be available to personalize the plan.

Main Flow:

37. The user opens the Learning Plan section.

38. The user selects a goal, such as improving speaking, grammar, or vocabulary, and provides any requested preferences.
39. The AI analyzes the available level, assessment results, strengths, and weaknesses.
40. The AI generates recommended activities, practice scenarios, and daily or weekly goals.
41. The application displays the plan to the user.
42. The user follows the activities and marks tasks as completed, where supported.
43. The application saves the plan and tracks completed tasks.

Alternative Flow:

If there is insufficient information, the application asks the user to complete an assessment or provide more details before generating a plan.

Postconditions:

A personalized plan is saved and available for the user to follow.

UC-09 – Update a Personalized Learning Plan

Actor: Learner and AI coach

Goal: Keep the plan aligned with current performance

Preconditions: A personalized learning plan and new performance data are available.

Main Flow:

44. The application obtains recent assessment results or conversation evaluations.
45. The AI reviews the user's latest performance against the learning goals.
46. The AI recommends continuing, changing, or adding activities.
47. The application presents the proposed updates to the user.
48. The plan is updated according to the application's implemented confirmation and saving flow.

Alternative Flow:

If there is no new performance data, the current plan remains active until enough information is available.

Postconditions:

The learning plan reflects the latest available performance information.

UC-10 – Manage Profile and Settings

Actor: Learner

Goal: Maintain account details and preferences

Preconditions: The user is logged in.

Main Flow:

49. The user opens Profile or Settings.
50. The application displays available account information and preferences.
51. The user updates supported details or settings.
52. The application validates and saves the changes.

Alternative Flow:

If the changes are invalid or cannot be saved, the application displays an error and retains the last saved information.

Postconditions: Supported profile details and preferences are updated.

4. End-to-End User Scenarios

Scenario A – First-Time Learner

53. The user creates an account and signs in.
54. The user selects an English level or completes the AI level assessment.
55. The user chooses a real-world scenario and practices with the AI coach.
56. The user receives an evaluation and reviews corrections.
57. The user saves useful vocabulary and starts building a learning history.

Scenario B – Checking Improvement

58. The user completes an initial assessment and saves the result.
59. The user practices regularly and reviews feedback.
60. After a period of learning, the user takes another assessment.
61. The application compares available results and highlights changes.
62. The user sees which skills improved and which still need practice.

Scenario C – Following a Personalized Plan

63. The user shares a learning goal and completes an assessment if needed.
64. The AI creates a plan with practice activities and learning goals.
65. The user completes tasks and practices relevant scenarios.
66. The application records available activity and performance data.
67. The AI recommends plan updates based on new results.